

Recognize situations that will result in any of the following being thrown: `ArrayIndexOutOfBoundsException`, `ClassCastException`, `IllegalArgumentException`, `IllegalStateException`, `NullPointerException`, `NumberFormatException`, `AssertionError`, `ExceptionInInitializerError`, `StackOverflowError` or `NoClassDefFoundError`. Understand which of these are thrown by the virtual machine and recognize situations in which others should be thrown programmatically.

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`ArrayIndexOutOfBoundsException`

Thrown to indicate that an array has been accessed with an illegal index. The index is either negative or greater than or equal to the size of the array.

```
java.lang.Object
  java.lang.Throwable
    java.lang.Exception
      java.lang.RuntimeException
        java.lang.IndexOutOfBoundsException
          java.lang.ArrayIndexOutOfBoundsException
```

`ClassCastException`

Thrown to indicate that the code has attempted to cast an object to a subclass of which it is not an instance. For example, the following code generates a `ClassCastException`:

```
Object x = new Integer(0);
System.out.println((String)x);
```

```
java.lang.Object
  java.lang.Throwable
    java.lang.Exception
      java.lang.RuntimeException
        java.lang.ClassCastException
```

`IllegalArgumentException`

Thrown to indicate that a method has been passed an illegal or inappropriate argument.

```
java.lang.Object
  java.lang.Throwable
    java.lang.Exception
      java.lang.RuntimeException
        java.lang.IllegalArgumentException
```

`IllegalStateException`

Signals that a method has been invoked at an illegal or inappropriate time. In other words, the Java environment or Java application is not in an appropriate state for the requested operation.

```
java.lang.Object
  java.lang.Throwable
    java.lang.Exception
      java.lang.RuntimeException
        java.lang.IllegalStateException
```

NullPointerException

Thrown when an application attempts to use null in a case where an object is required. These include:

- Calling the instance method of a null object.
- Accessing or modifying the field of a null object.
- Taking the length of null as if it were an array.
- Accessing or modifying the slots of null as if it were an array.
- Throwing null as if it were a Throwable value.
- Applications should throw instances of this class to indicate other illegal uses of the null object.

```
java.lang.Object
  java.lang.Throwable
    java.lang.Exception
      java.lang.RuntimeException
        java.lang.NullPointerException
```

NumberFormatException

Thrown to indicate that the application has attempted to convert a string to one of the numeric types, but that the string does not have the appropriate format.

```
java.lang.Object
  java.lang.Throwable
    java.lang.Exception
      java.lang.RuntimeException
        java.lang.IllegalArgumentException
          java.lang.NumberFormatException
```

AssertionError

Thrown to indicate that an assertion has failed. The seven one-argument public constructors provided by this class ensure that the assertion error returned by the invocation:

```
new AssertionError(expression)
```

```
java.lang.Object
  java.lang.Throwable
    java.lang.Error
      java.lang.AssertionError
```

ExceptionInInitializerError

Signals that an unexpected exception has occurred in a static initializer. An `ExceptionInInitializerError` is thrown to indicate that an exception occurred during evaluation of a static initializer or the initializer for a static variable.

As of release 1.4, this exception has been retrofitted to conform to the general purpose exception-chaining mechanism. The "saved throwable object" that may be provided at construction time and accessed via the `getException()` method is now known as the cause, and may be accessed via the `Throwable.getCause()` method, as well as the aforementioned "legacy method."

```
java.lang.Object
  java.lang.Throwable
    java.lang.Error
      java.lang.LinkageError
        java.lang.ExceptionInInitializerError
```

StackOverflowError

Thrown when a stack overflow occurs because an application recurses too deeply.

```
java.lang.Object
  java.lang.Throwable
    java.lang.Error
      java.lang.VirtualMachineError
        java.lang.StackOverflowError
```

NoClassDefFoundError

Thrown if the Java Virtual Machine or a `ClassLoader` instance tries to load in the definition of a class (as part of a normal method call or as part of creating a new instance using the `new` expression) and no definition of the class could be found.

The searched-for class definition existed when the currently executing class was compiled, but the definition can no longer be found.

```
java.lang.Object
  java.lang.Throwable
    java.lang.Error
      java.lang.LinkageError
        java.lang.NoClassDefFoundError
```

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Recognize the effect of an exception arising at a specified point in a code fragment. Note that the exception may be a runtime exception, a checked exception, or an error.

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